

OCR-Aware Question Solver: Comprehensive Engineering Write-Up

Retry Cap & Refinement Discipline The maximum retry cap for the OCR correction loop is strictly set to **3**. This decision balances accuracy with system efficiency. Empirical testing indicates that multimodal LLMs generally identify and fix obvious visual discrepancies (such as misread exponents, broken matrices, or swapped digits) within the first or second refinement pass. If the agent cannot find a matching option after 3 attempts, it strongly implies that the image crop is fundamentally flawed, the question itself contains an inherent logical error, or the provided multiple-choice options are incorrect. Extending the loop beyond 3 attempts yields diminishing returns, drastically increases the risk of LLM hallucinations (where the model starts rewriting the core logic of the question to force a match), and unnecessarily inflates API token costs and processing latency.

Handling Unresolved Cases (Best-Guess Fallback) When the system exhausts its retry cap without successfully matching an option, it must still deliver a definitive output. To determine the “best-guess answer,” the agent analyzes the reasoning chain from its final attempt:

- For **quantitative or mathematical questions**, the solver extracts the final derived numerical value. It then calculates the absolute difference between this value and the available numeric options, selecting the option that is mathematically closest.
- For **qualitative or text-based questions**, the agent utilizes a semantic similarity heuristic (such as string overlap or conceptual mapping) to align its derived conclusion with the most relevant option.

The final output continues to conform to the strict JSON schema, returning the chosen best-guess option, ensuring the pipeline remains unbroken even when a perfect match is impossible.

Synthetic Noise Perturbation Strategy To rigorously exercise the agent’s self-healing capabilities, a synthetic noise injector introduces realistic Persian OCR degradations. The perturbation rate is explicitly capped at **5 to 7 characters or symbols per question block**. This specific threshold is deliberate: it introduces enough friction to reliably trigger the correction loop by swapping visually confusable elements (e.g., Persian digits ۲/۳, ۴/۶, or letters گ/ک, پ/ب). This ensures the text is realistically degraded without obliterating the fundamental mathematical or textual context of the question. Injecting too much noise turns the task into an impossible decryption problem, while this targeted degradation accurately mimics the artifacting seen in low-resolution scans of actual Iranian exam papers.